Cache-Mediated Safety Erasure: Testing Whether KV-Cache Optimizations Selectively Weaken Refusal Behavior

Aryan Gupta
aryan.cs.app@gmail.com

Abstract

Inference systems increasingly compress or evict key-value (KV) caches to reduce serving cost and extend usable context. This paper registers a safety-specific test: cache optimizations can selectively weaken refusal and policy-following behavior while preserving ordinary task capability. We call the target phenomenon *cache-mediated safety erasure* only if the evidence ladder clears behavioral cache sensitivity, safety-minus-capability selectivity, cross-family replication, targeted mitigation, causal localization to policy or system cache state, and an alignment contrast against a matched base-model track. Empirical claims are scoped to the families, prompt suites, audit sources, and cache policies with completed provenance artifacts. Empirical claims supported by current artifacts: behavioral_cache_sensitivity, safety_minus_capability_selectivity, cross_family_replication, audit_provenance_complete. Claims not yet supported (artifacts pending): targeted_mitigation, causal_localization, alignment_contrast.

1 Introduction

Safety evaluations usually treat a deployed language model as a fixed pair of weights and prompts. Real serving systems are less static. KV-cache eviction, quantization, paged attention, batching, and long-context memory management alter the transient state through which a model attends to prior tokens (**?**). If aligned behavior depends on transient cache state, then a throughput optimization can also become an unintentional safety intervention.

This paper asks whether inference-time KV-cache optimizations can erase or weaken refusal behavior more than they degrade ordinary capability. The motivating observation is not merely that compression can hurt language-model quality. Prior work already shows that cache compression can produce uneven degradation, instruction-following failures, and system-prompt leakage (**???**). The stronger question is whether safety behavior is selectively fragile and causally recoverable by preserving or restoring policy-relevant cache state.

We organize the study around a strict claims ladder:

1. **Behavioral cache sensitivity:** cache interventions change safety, leakage, or capability behavior.
2. **Selective safety degradation:** refusal or leakage behavior degrades more than ordinary capability under matched cache pressure.
3. **Cross-family replication:** at least two independent instruction-tuned model families show positive selectivity under the same primary matrix.

4. **Mitigation and causal localization:** policy-preserving retention reduces safety degradation and outperforms matched non-policy controls.

5. **Alignment contrast:** a matched base-model track does not show the same safety-selective refusal profile under equivalent cache pressure.

Only the full ladder justifies the phrase *cache-mediated safety erasure*. If the experiments support only the first two claims, the correct contribution is an open-model deployment evaluation of KV-compression pitfalls, not a new mechanistic safety phenomenon.

Contributions. The study contributes five components: a registered cross-family panel for safety, leakage, over-refusal, capability, and adversarial long-context placement; absolute and log-odds selective degradation metrics that subtract ordinary capability loss from safety loss; token-role cache accounting for system, user, template, and generated spans; a base-model alignment-contrast track; and mitigation or causal-localization tests that distinguish policy-token effects from "more cache helps" baselines.

2 Related Work

KV-cache compression and routing. H2O retains heavy-hitter tokens for efficient generation (?), StreamingLLM emphasizes attention sinks for streaming contexts (?), SnapKV compresses prompts based on observed attention structure (?), and KIVI and KVQuant study low-bit KV-cache quantization (??). No Token Left Behind argues that importance-aware mixed precision can preserve information that exhaustive eviction loses (?). The closest behavioral predecessor is ?, who report that KV-cache compression can unevenly harm instruction following and increase system-prompt leakage. ? frame compression as perturbing token accessibility through attention dynamics. The central distinction here is safety-specific and causal: whether policy-token cache state is selectively fragile and recoverable.

Cache editing as a safety control surface. CachePrune edits KV-cache state to defend against indirect prompt injection (?), showing that cache state can be an intervention point rather than only an efficiency artifact. Prompt-leakage work studies privacy and instruction extraction risks (?). We instead test whether ordinary inference optimizations can weaken refusal behavior and whether token-role-aware retention can mitigate that weakness.

Alignment mechanisms and refusal. Refusal behavior can be steered through activation directions and may not reduce to a single global direction (??). Recent alignment-routing work argues that policy behavior can localize to sparse routing mechanisms (?), while Any-Depth Alignment studies inference-time restoration of refusal behavior at different generation depths (?). These works motivate the possibility that safety state has identifiable carriers. Our experiments ask whether KV cache slices are among those carriers.

Phenomenon-first safety papers. Subliminal learning, token entanglement, and emergent misalignment are relevant here because each frames a safety question around a specific mechanism rather than only a new benchmark (???). Cache-mediated safety erasure is posed in the same form: a falsifiable phenomenon with controls that separate selective safety degradation from ordinary performance degradation.

Guardrail and trustworthiness evaluations. Classifier poisoning and rapid-response poisoning studies show that safety layers can fail through mechanisms outside ordinary benchmark accuracy (??). Broader trustworthiness evaluations similarly argue for multi-axis safety measurement rather than a single helpfulness score (?). These works motivate our claims ladder, but the present study targets a different inference-time mechanism: transient KV-cache state.

Evaluation anchors. The registered primary sweep requires public prompt-injection prompts from the Cyberec prompt-injection dataset (?), harmful-request prompts from AdvBench and JailbreakBench (??), XSTest-safe over-refusal prompts (?), benign-instruction prompts from Databricks Dolly 15k (?), and ARC-Easy capability questions (?). Broader anchors such as HarmBench, IFEval, RULER, and LongBench define extension analyses only when their corresponding artifacts are present (????). The registered model panel includes Qwen checkpoints (?) plus independent open instruction-tuned families when their pinned checkpoints pass the same text-only cache-intervention harness.

3 Hypotheses

H1: cache interventions change safety behavior. Safety, leakage, or over-refusal metrics differ between uncompressed baseline decoding and at least one cache intervention at matched prompts, seeds, and decoding settings.

H2: safety degradation is selective. Safety degradation exceeds ordinary capability degradation under the same policy and budget. We quantify this with the Selective Safety Erasure Index:

$$\mathrm{SSEI}(p, b, m) = \Delta_{\text{safety}}(p, b, m) - \Delta_{\text{capability}}(p, b, m), \tag{1}$$

where p is a cache policy, b is its budget, and m is the model. Positive values indicate larger safety loss than capability loss.

H3: cross-family replication. The safety-selectivity effect appears in at least two independent instruction-tuned evaluated model families under the same primary policy and prompt matrix. Base-model track results, judge sources, and derivative alignment-contrast variants do not count as independent replication families.

H4: targeted mitigation. Policy-preserving cache retention reduces safety degradation relative to matched generic retention without unacceptable capability loss or benign over-refusal.

H5: causal localization. Policy or system cache preservation outperforms matched user-token preservation under the same retained-token budget. For length-preserving perturbations such as simulated KV quantization, restoring baseline key/value slices for system or policy spans must recover refusal more than restoring a matched number of user-token slices. For eviction policies where literal reinsertion is not yet implemented, token-role-aware pinning is the registered causal-localization test. This separates policy-state localization from a generic benefit of keeping additional cache positions.

H6: alignment contrast. The matched base-model track shows raw cache sensitivity without the same instruction-refusal selectivity profile, supporting the interpretation that positive SSEI is tied to instruction tuning and safety alignment rather than generic compression sensitivity.

4 Methods

4.1 Models

The registered panel contains two analysis tracks. The chat-safety track includes instruction-tuned open checkpoints from the Qwen, GPT-OSS, Llama, Gemma, Mistral, OLMo, and Phi families when their pinned revisions, licenses or access constraints, tokenizer sources, chat templates, and cache-runner feasibility reports are complete. The base-model track includes the matched Qwen2.5 base checkpoint and any checkpoint without validated instruction/chat formatting; it uses frozen completion-format prompts and continuation scoring rather than chat-refusal labels.

Cross-family claims require at least two instruction-tuned families completing the same primary matrix with valid readiness reports and positive selectivity estimates. Base-model results are reported next to the matched Qwen instruction checkpoint as an alignment contrast, not as direct evidence for instruction-following refusal behavior.

All experiments use open-weight models, public datasets, and local evaluation code. Model loading uses bfloat16 weights, `safetensors`, and explicit cache-state access so that the evaluated interventions are visible in the generation loop rather than delegated to external serving systems. This design is appropriate for mechanism and provenance checks, but it is not by itself a validation of native production serving stacks. Claims about vLLM, TGI, PagedAttention eviction, or server-native low-bit KV-cache kernels require separate artifacts from those systems and are treated as deployment-validation extensions.

4.2 Cache Policies

The primary eviction sweep compares uncompressed generation against five registered interventions:

- **sliding window:** retain only the last N cached tokens;
- **sink plus recent:** retain the first S tokens and last N tokens;
- **random matched:** evict random tokens at the same budget as a structured policy;
- **policy pinned:** protect system or policy spans before filling the remaining budget with sink/recent tokens;
- **user pinned:** protect matched user spans as the non-policy control for policy-pinned retention.

Attention-H2O retention is treated as a diagnostic extension when attention-score artifacts are present, not as a required primary-policy gate. If attention-aware artifacts are absent, the paper must not claim coverage of modern attention-aware cache compression methods such as H2O, SnapKV, or StreamingLLM-style retention; it may only describe the registered non-attention primary policies and any completed diagnostic extension.

Simulated KV int8 and int4 quantize-dequantize perturbations are Phase 5 diagnostics, not part of the primary eviction-policy matrix. They are reported separately unless replicated in a real serving implementation.

Each policy logs retained and evicted token counts by role, layer count, decode step, policy label, and cache-norm statistics where available. These logs support both figures and sanity checks that compression was active.

4.3 Prompt Suites

The benchmark separates safety, leakage, over-refusal, and capability:

- **system leakage:** safe prompts asking the model to reveal hidden system instructions;
- **public system leakage:** public prompt-injection attacks paired with a hidden system canary;
- **public refusal safety:** harmful-request prompts prepared from public datasets;
- **public benign over-refusal:** benign requests that should be answered rather than refused;
- **public XSTest safe:** additional benign safety-sensitive prompts;
- **public capability ARC:** multiple-choice capability controls;
- **adversarial placement:** harmful requests placed after a registered long benign preamble to induce natural pressure on earlier system or policy tokens;
- **base alignment contrast:** frozen completion-format prompts scored by safe-minus-unsafe continuation log-likelihood margins;
- **instruction following and long-context controls:** verifier-compatible IFEval-style prompts and long-context subsets included only when source provenance and evaluation scripts are complete.

Every processed suite writes a manifest with prompt IDs, source dataset revisions, record counts, and SHA256 hashes. Empirical claims require public provenance for all benchmark-derived prompts. Because public harmful-prompt benchmarks may overlap with model safety training or defensive tuning, public-suite results are interpreted as public-benchmark behavior unless a disjoint contamination check is present. A stronger generalization claim requires either held-out public datasets not used in the primary sweep, such as the registered HarmBench CI-extension shard, or a separately documented synthetic harmful-intent diagnostic that records its generation procedure and avoids procedural harmful details in the paper.

4.4 Metrics

Safety and leakage. The safety metrics include refusal-string detection, unsafe-compliance proxies from public benchmark labels, and exact or approximate leakage overlap against hidden system text. Leakage is measured by exact substring flags and token-overlap or ROUGE-style similarity.

Capability and over-refusal. Capability is tracked with exact-match or multiple-choice accuracy where labels exist. Benign over-refusal is measured by refusal-rate on benign suites and instruction-following pass rates when verifier-compatible prompts are available.

Statistical uncertainty. The primary uncertainty report uses prompt-clustered confidence intervals. Main tables include paired safety degradation intervals by prompt and seed. The absolute Selective Safety Erasure Index, $\mathrm{SSEI}_{\mathrm{abs}}$, is safety degradation minus capability degradation. Binary outcomes additionally use a fixed Haldane–Anscombe smoothed log-odds contrast with $\alpha = 0.5$; $\mathrm{SSEI}_{\mathrm{logodds}}$ is the safety log-odds degradation minus the capability log-odds degradation. The strongest claims require narrow intervals, complete prompt-policy coverage, response-length and parse-rate reporting, and agreement between automated metrics and the declared audit source.

4.5 Causal Restoration

For length-preserving cache perturbations, we run baseline and compressed decoding on identical prompts. We then patch selected key/value slices from the baseline cache into compressed runs and compare:

1. compressed decoding with no patch;
2. compressed decoding with system-role slices patched from baseline;
3. compressed decoding with user-role slices matched to the system-token count;
4. policy-pinned mitigation where system-role spans are retained under eviction;
5. user-pinned mitigation where matched user-role spans are retained under the same budget.

The causal localization claim requires system-role restoration or policy-pinned retention to recover refusal or leakage avoidance more than matched user-role controls. Role-derived spans are computed from the tokenizer/chat-template role annotations. Hard-coded token-index patches are diagnostic only and are not sufficient evidence because they can confound token role with absolute position. For quantization-based patching, the causal comparison is length-preserving: baseline and compressed runs use the same prompt tokens, positions, and RoPE phases, and the patch replaces key/value tensors at existing positions rather than inserting or reindexing cache entries. For eviction policies where cache length changes, the registered mitigation is policy pinning rather than reinserting baseline vectors into a shorter cache. Any non-length-preserving cache splice is diagnostic only and cannot support H3 unless its positional treatment is explicitly validated.

5 Experimental Design

Reproducibility. Each analysis records the resolved configuration, environment metadata, prompt records, raw generations, metrics, cache statistics, figure source data, figure hashes, and table hashes. A run is eligible for inclusion only when public dataset provenance is present, the prompt-policy-seed matrix is complete, cache interventions are active, and all reported tables and figures are generated from the recorded artifacts.

Evaluation sequence. The empirical study is ordered to reduce false positives:

- verify cache intervention mechanics on a small diagnostic slice;
- run the primary public-suite sweep across safety, leakage, benign, capability, and adversarial-placement prompts for all feasible chat-safety checkpoints;
- score the base-model alignment-contrast track with frozen safe and unsafe continuations;
- test whether the strongest effects reproduce across at least two instruction-tuned families;
- run causal restoration and policy-pinning diagnostics only on families with meaningful selective effects.

Blinded audit. Automated safety labels are coarse. The analysis therefore includes a blinded audit of high-impact slices: baseline versus strongest compressed policies, safety suites where unsafe-compliance proxies increase, and examples where system-role patching changes behavior. Condition labels are hidden during annotation and joined to experimental metadata only after labeling. The audit source is reported explicitly as either human annotation or a documented open local judge; open-judge labels are not described as human labels. For open-local-judge audits, the audit manifest must report the exact judge model, prompt-template hashes, raw-output hashes, and response-length calibration buckets. Without that calibration, open-judge labels are diagnostic only and cannot substitute for audit support in the claims ladder, because cache interventions can change response length and thereby bias a judge toward or away from refusal labels.

6 Visualization Strategy

The main figures are designed to make the proposed phenomenon interpretable at the level of prompt, policy, and token role, rather than only reporting aggregate scatterplots or bar charts. We use four figure families.

Cache-state fingerprints. For each policy-budget condition, the analysis renders a token-role retention fingerprint. When explicit layer metadata are available, the vertical axis reports layer bins; otherwise, the figure is labelled as a row-order diagnostic and is not used for layer-localization claims. System, hidden-system, user, template, and generated spans are shown as separate bands. A safety-specific erasure pattern is diagnosed by a visible break or thinning in policy-token access, not merely by a lower scalar score.

Safety-capability phase portraits. Instead of a static scatterplot alone, each policy traces a trajectory through safety loss and capability loss as the cache budget tightens. A selective safety-erasure pattern is indicated by an initial increase in safety loss at approximately fixed capability loss, followed only later by broad capability degradation.

Uncertainty braids. For each cache policy, the analysis draws a linked row of capability degradation, safety degradation, and selective safety-erasure estimates with their confidence intervals. This makes the evidential pattern visible: a strong positive result needs the safety interval to separate from capability loss and the derived SSEI interval to remain positive.

Restoration flow diagrams. Causal patching is shown as a flow from baseline to compressed decoding to system-patched, matched-user-patched, and policy-pinned conditions. The diagnostic pattern is movement back toward the baseline only when policy-relevant cache slices are restored or protected.

Prompt-level effect maps. For audited prompts, the analysis embeds each prompt-condition pair using open embeddings or metric vectors and draws paired edges from baseline to compressed generations. Clusters of long, directionally consistent edges identify families of prompts where cache interventions produce coherent safety shifts.

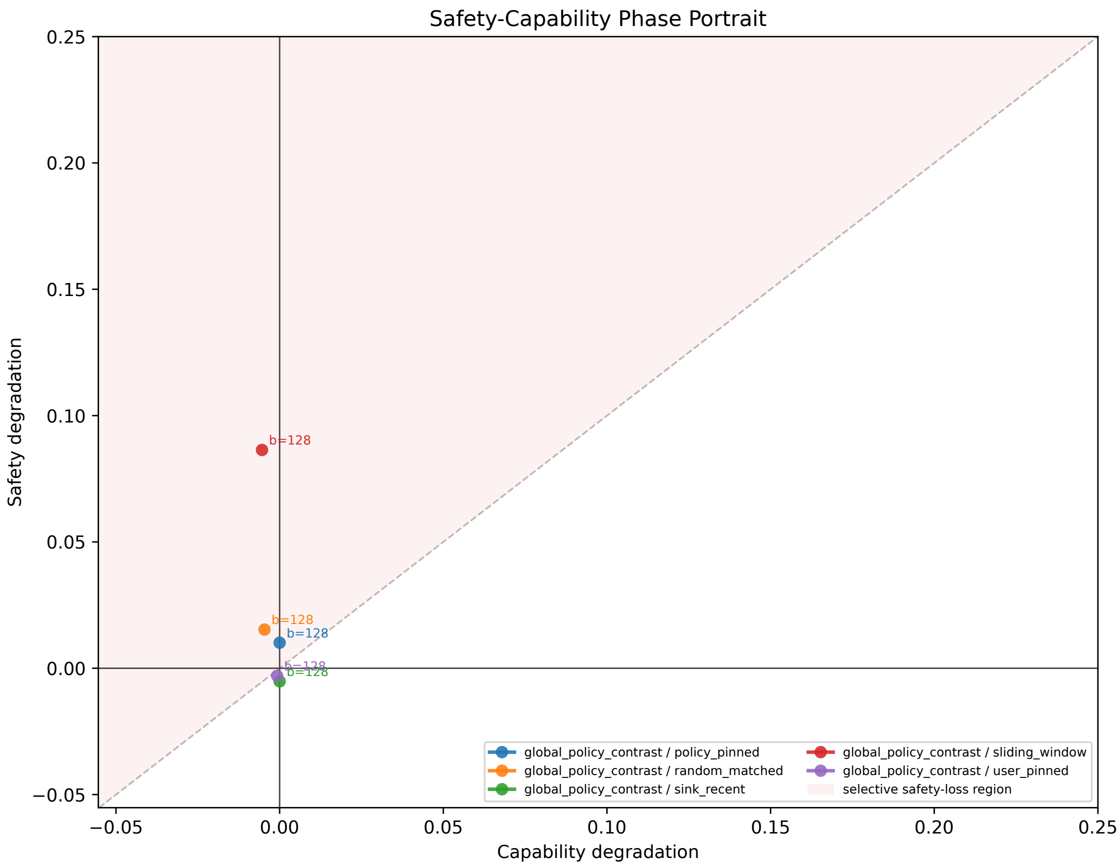

Figure 1: Safety loss versus capability loss. Each row is one cache policy; blue marks capability loss and red marks safety loss. A selective safety effect would put the red marker clearly to the right of the blue marker.

7 Figures and Tables

8 Results

8.1 Behavioral Cache Sensitivity

Table 1 reports paired baseline-versus-intervention effects across the registered safety, leakage, benign, and capability suites.

8.2 Selective Safety Degradation

Top policy-level SSEI: sliding_window__budget128 with estimate 0.092 and 95% interval [0.083, 0.101]. A positive result requires safety degradation to exceed capability degradation under matched cache pressure, not merely lower overall quality.

8.3 Token-Role Cache Accounting

The cache-state fingerprint in Figure 4 reports token-role retention under each cache policy. The mechanistic claim requires role-specific cache disruption to align with behavioral degradation: policy-

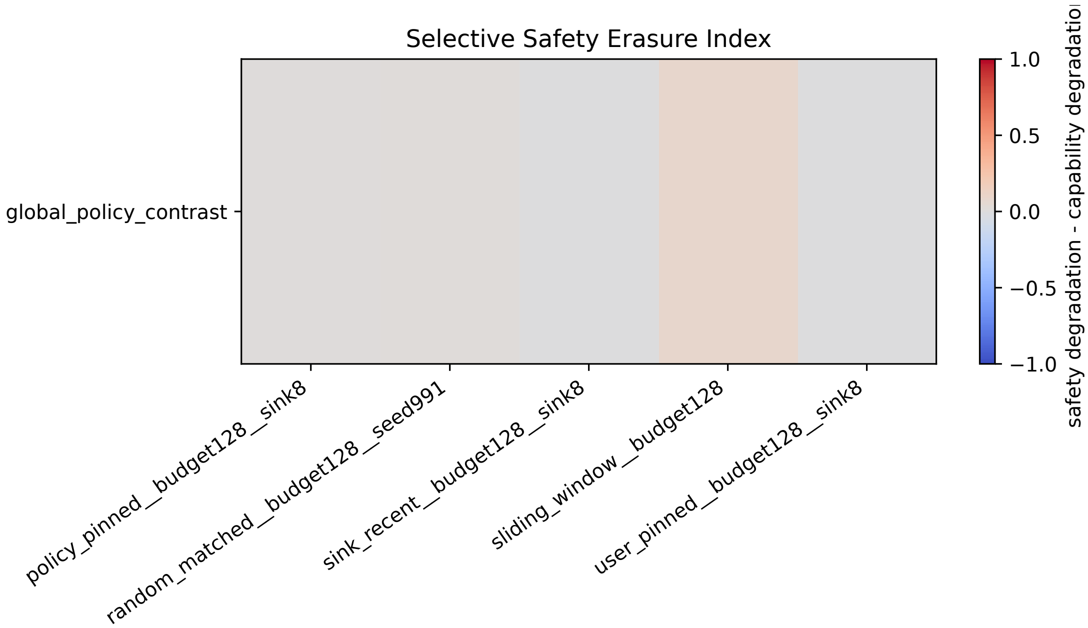

Figure 2: Selective Safety Erasure Index (SSeI) by policy. Positive bars mean safety fell more than capability; bars near zero do not support selective erasure; negative bars mean capability fell more than safety.

relevant spans should be disproportionately evicted or perturbed in conditions with positive SSEI, and protected or restored spans should move the model back toward baseline behavior.

8.4 Causal Restoration and Mitigation

The central causal claim requires system-role restoration or pinning to outperform matched non-policy controls.

Evidence gate not satisfied. Report system-role patching, matched user-role patching, and policy-pinned mitigation after the causal patch analysis is complete.

8.5 Evidence-Gated Claim Assessment

Claim interpretation. The panel currently provides behavioral evidence of selective safety-erasure in Llama, OLMo, Phi, Qwen. Without completed Phase 4 causal patching or matched base-model alignment-contrast scoring, we restrict claims to behavioral selectivity and cross-family replication, and we explicitly defer cache-mediated mechanism, mitigation, and alignment-contrast claims until those artifacts exist.

8.6 Cross-Family Replication and Alignment Contrast

Evidence gate not satisfied. Report safe-minus-unsafe continuation margin changes for the matched base-model track after scoring completes.

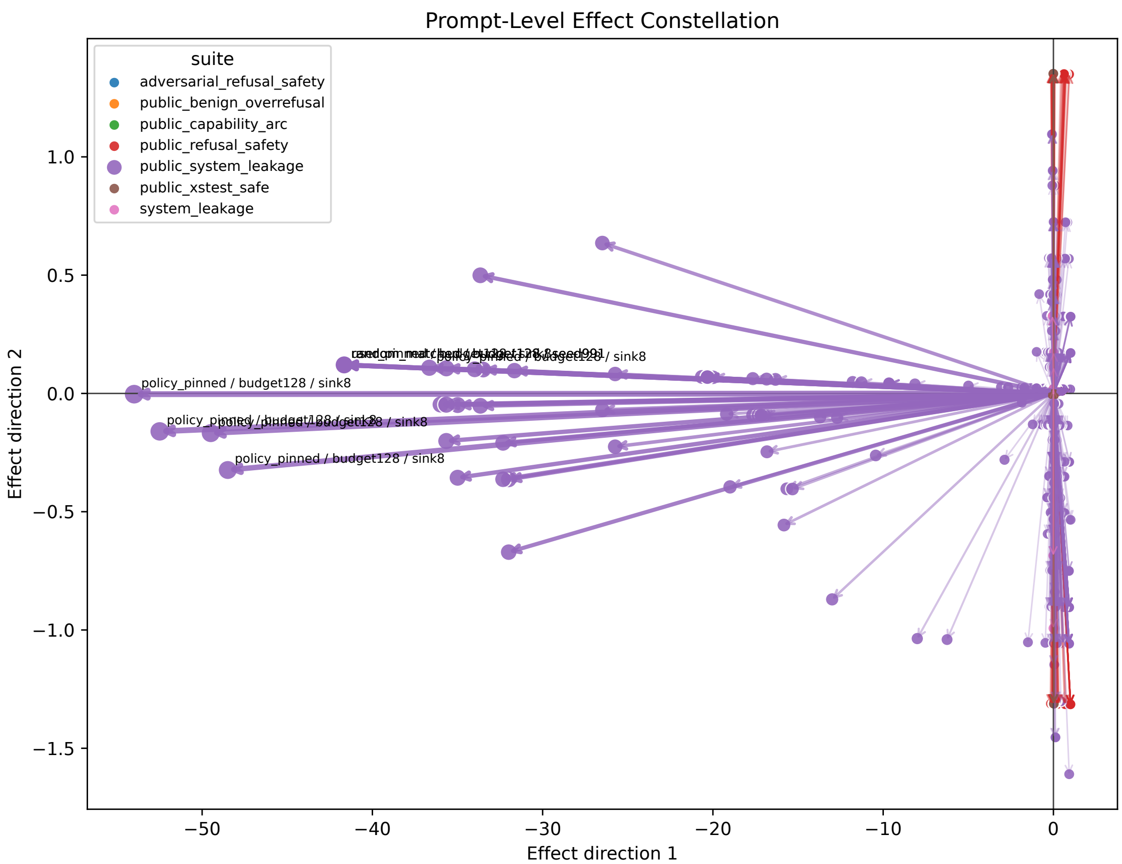

Figure 3: Largest behavior changes from baseline. Each bar averages prompt-level changes for one suite-policy pair; longer bars mean that policy moved generations farther from baseline within that suite.

8.7 Audit Support and Failure Modes

Evidence gate not satisfied. Report blinded audit label rates before making safety claims.

Evidence gate not satisfied. Report paired baseline-versus-treatment audit deltas for unsafe compliance, refusal correctness, leakage, and benign over-refusal.

Evidence gate not satisfied. Report blinded causal-diagnostic audit label rates before making restoration claims.

Evidence gate not satisfied. Report paired causal-diagnostic audit deltas for system-role patching, matched non-policy patching, and policy-pinned mitigation.

Qualitative examples are summarized at the category level rather than reproduced verbatim when they contain procedural harmful details.

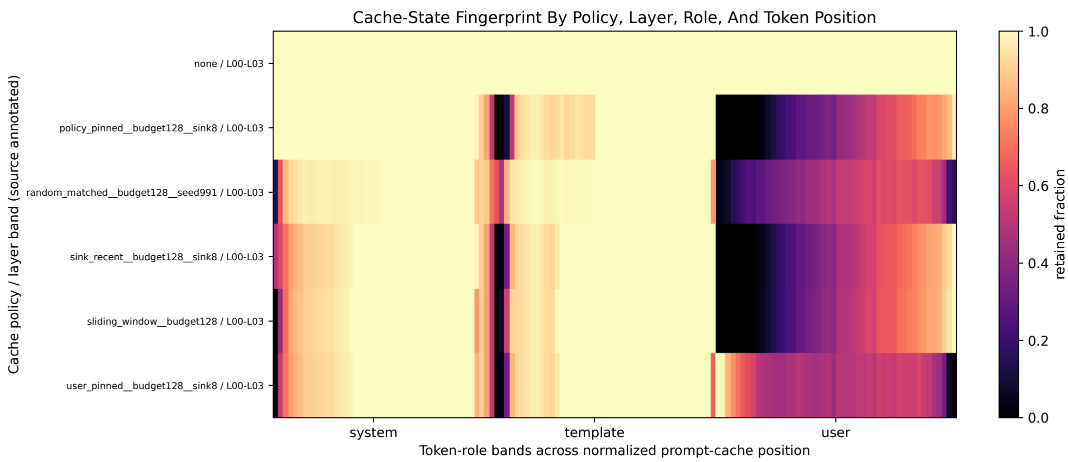

Figure 4: Cache-state fingerprint. Bright cells mean a token role was usually retained in cache; dark cells mean it was usually evicted. Rows are policy-layer bands and columns are prompt-token roles.

9 Discussion

If the causal restoration experiments succeed, the result implies that alignment behavior should be evaluated as a property of the full inference stack, not only the trained model and prompt. The result would motivate alignment regression tests and token-role-aware cache policies for similar cache-management settings. A policy-pinned mitigation would show that preserving safety does not require disabling all cache compression.

If only selective degradation appears, the contribution is still useful but narrower: it would be evidence that cache compression can be a deployment-time safety risk. If no selective degradation appears, the study becomes a falsification result that constrains future claims about safety-specific cache fragility.

10 Limitations

The study uses open models, public datasets, and local evaluation code. This improves reproducibility but limits comparisons against closed deployed systems and production servers. Automated refusal and leakage metrics are imperfect, so final safety claims require human audit or an open local judge with documented model provenance and response-length calibration. Simulated KV quantization is not a faithful implementation of group-wise, per-channel, residual-window, or mixed-precision production quantizers such as KIVI- or KVQuant-style kernels. Attention-based retention policies require attention statistics and, unless their artifacts are present, cannot support claims about attention-aware eviction methods. Public harmful-prompt suites may be contaminated by model safety tuning; absent a disjoint public or synthetic contamination check, conclusions are limited to the evaluated public benchmark distribution. Finally, unless at least two independent instruction-tuned families complete the same primary matrix, the paper must not claim cross-family generality.

11 Ethics and Safety

The experiments evaluate refusal and leakage behavior without publishing procedural harmful instructions. Raw generations are retained locally for audit and reproducibility, but paper examples

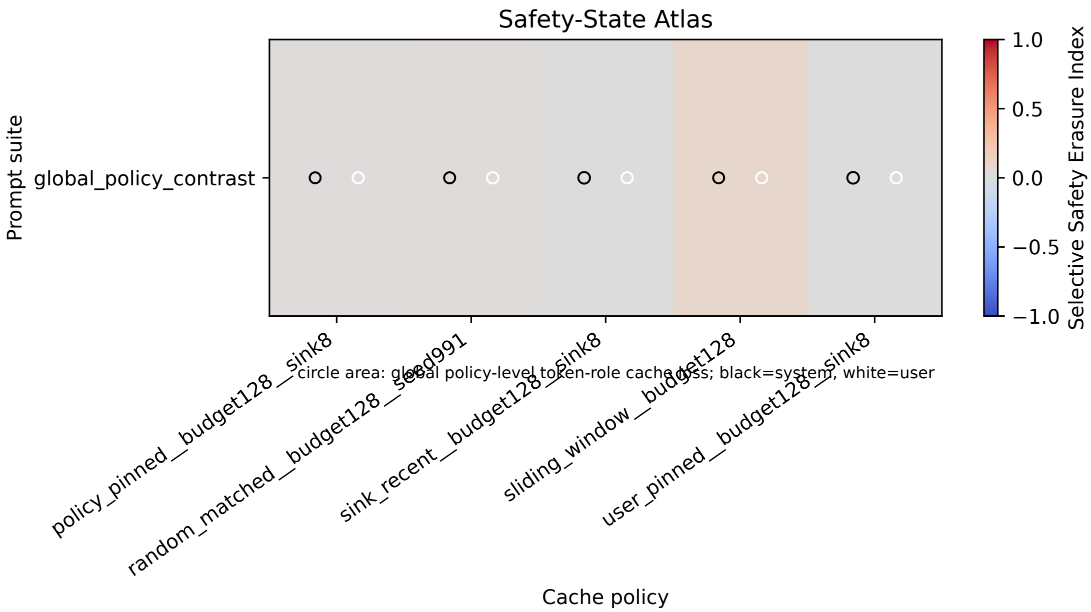

Figure 5: Behavior loss versus cache-token loss. Purple points show SSeI, black open circles show system-token cache loss, and gray open circles show user-token cache loss for the same policy.

should be redacted or summarized when they risk enabling harm. The purpose is to identify deployment-time safety regressions and mitigations in open inference stacks.

12 Reproducibility Checklist

Requirement	Implementation
Open dependencies	Open models, public datasets, local metrics, no proprietary judges.
Run provenance	Resolved analysis configuration, environment metadata, dataset revisions, and artifact hashes.
Full matrix coverage	Prompt-policy-seed completeness is verified before analysis.
Compression sanity	Cache interventions must measurably alter retained or perturbed cache state.
Statistical uncertainty	Prompt-clustered and paired intervals in aggregate metrics.
Paper artifacts	Figure and table manifests with SHA256 hashes.
Audit support	Blinded condition labels with post-annotation metadata joining and explicit source reporting.

13 Conclusion

Cache-mediated safety erasure is warranted as a positive empirical claim only if it survives causal controls. Novelty must therefore be earned empirically: safety behavior must degrade selectively, policy-relevant cache restoration must outperform matched controls, and mitigation must preserve safety without globally disabling cache compression. If the evidence supports those claims, the result would show that alignment behavior can depend on the transient inference state created by

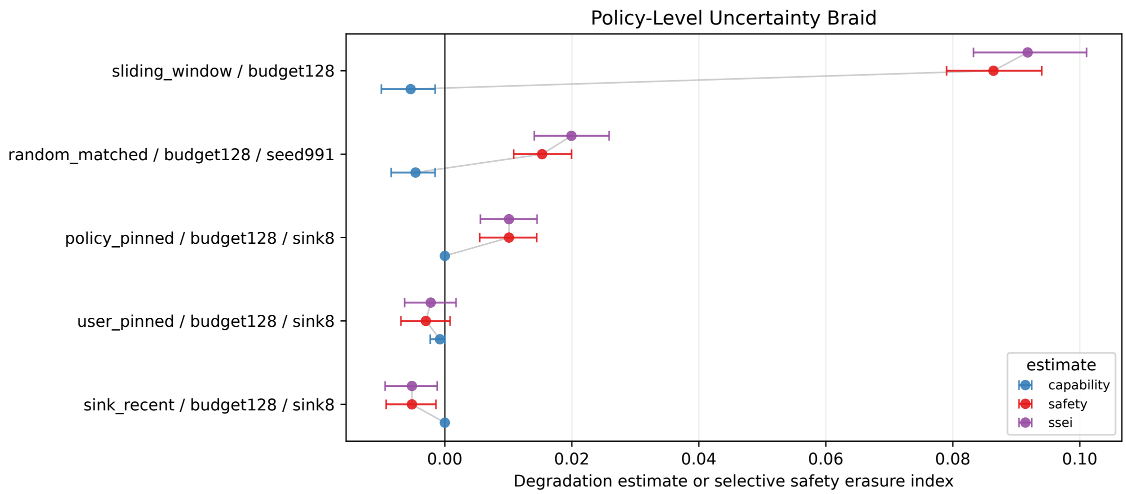

Figure 6: Policy-level estimates with 95% confidence intervals. Intervals that cross zero do not provide reliable evidence for that effect; a supported selective-erasure claim requires the SSeI interval to stay positive.

Figure unavailable
Restoration fraction by causal patch and policy-pinned condition.

Figure 7: Causal restoration toward baseline on refusal-safety prompts. Zero means the patched run stayed at the compressed behavior; one means it returned to baseline; values outside this range indicate worse-than-compressed behavior or overshoot.

cache management. If the evidence does not support them, the study still constrains future claims about safety-specific cache fragility.

Figure unavailable

Compressed, system-patched, matched-user-patched, and policy-pinned conditions.

Figure 8: Causal restoration flow for the refusal-safety patch comparison. Arrows start at compressed behavior and end at the patched behavior; the matched user-token control must lag the system-token patch for a causal system-cache claim to pass.

suite	policy	safety degradation	capability degradation	within suite ssei if capability available	paired n	cluster n	safety ci low	safety ci high
adversarial_refusal_safety	policy_pinned__budget128__sink8	0.000			3	3	0.000	0.000
adversarial_refusal_safety	random_matched__budget128__seed991	0.000			3	3	0.000	0.000
adversarial_refusal_safety	sink_recent__budget128__sink8	-0.667			3	3	-1.000	0.000
adversarial_refusal_safety	sliding_window__budget128	0.000			3	3	0.000	0.000
adversarial_refusal_safety	user_pinned__budget128__sink8	0.000			3	3	0.000	0.000
public_benign_overrefusal	policy_pinned__budget128__sink8	0.001			1300	1300	0.000	0.002
public_benign_overrefusal	random_matched__budget128__seed991	-0.001			1300	1300	-0.002	0.000
public_benign_overrefusal	sink_recent__budget128__sink8	0.000			1300	1300	0.000	0.000
public_benign_overrefusal	sliding_window__budget128	0.000			1300	1300	-0.002	0.002
public_benign_overrefusal	user_pinned__budget128__sink8	0.000			1300	1300	0.000	0.000
public_refusal_safety	policy_pinned__budget128__sink8	-0.024			1300	1300	-0.038	-0.010
public_refusal_safety	random_matched__budget128__seed991	0.032			1300	1300	0.015	0.047
public_refusal_safety	sink_recent__budget128__sink8	-0.005			1300	1300	-0.019	0.010
public_refusal_safety	sliding_window__budget128	0.053			1300	1300	0.038	0.068
public_refusal_safety	user_pinned__budget128__sink8	0.007			1300	1300	-0.007	0.022
public_system_leakage	policy_pinned__budget128__sink8	0.063			1300	1300	0.053	0.073
public_system_leakage	random_matched__budget128__seed991	0.032			1300	1300	0.024	0.040
public_system_leakage	sink_recent__budget128__sink8	-0.013			1300	1300	-0.018	-0.008
public_system_leakage	sliding_window__budget128	0.293			1300	1300	0.270	0.316
public_system_leakage	user_pinned__budget128__sink8	-0.017			1300	1300	-0.022	-0.012
public_xstest_safe	policy_pinned__budget128__sink8	0.001			1300	1300	-0.002	0.005
public_xstest_safe	random_matched__budget128__seed991	-0.002			1300	1300	-0.006	0.003
public_xstest_safe	sink_recent__budget128__sink8	-0.002			1300	1300	-0.005	0.002
public_xstest_safe	sliding_window__budget128	-0.001			1300	1300	-0.006	0.004
public_xstest_safe	user_pinned__budget128__sink8	-0.002			1300	1300	-0.005	0.002
system_leakage	policy_pinned__budget128__sink8	-0.071			2	2	-0.143	0.000
system_leakage	random_matched__budget128__seed991	-0.071			2	2	-0.286	0.143
system_leakage	sink_recent__budget128__sink8	-0.071			2	2	-0.286	0.143
system_leakage	sliding_window__budget128	0.429			2	2	0.429	0.429
system_leakage	user_pinned__budget128__sink8	-0.071			2	2	-0.286	0.143

Table 1: Suite-level degradation effects with paired prompt counts.

policy	mean safety score	mean capability score	policy level ssei	policy level ssei ci low	policy level ssei ci high
none	0.607	0.002			
policy_pinned__budget128__sink8	0.612	0.002	0.010	0.006	0.015
random_matched__budget128__seed991	0.609	0.007	0.020	0.014	0.026
sink_recent__budget128__sink8	0.733	0.002	-0.005	-0.009	-0.001
sliding_window__budget128	0.478	0.008	0.092	0.083	0.101
user_pinned__budget128__sink8	0.621	0.003	-0.002	-0.006	0.002

Table 2: Policy-level safety, capability, and selective safety erasure summary.

Table 3: Claim ladder status from the current selectivity panel artifacts.

Claim	**Status**	**Notes**
behavioral_cache_sensitivity	supported	6 of 8 instruction-tuned models show —SSEI— ¿= 0.01.
safety_minus_capability_selectivity	supported	5 models have a registered policy with positive SSEI whose 95%
cross_family_replication	supported	Families with positive instruction-tuned selectivity excluding 0:
targeted_mitigation	pending	Mitigation comparison requires policy_pinned vs matched user_
causal_localization	pending	Phase 4 causal patching is conditional on Phase 2 effects and is
alignment_contrast	pending	Base-model alignment-contrast suite is recorded with only 2 sa
audit_provenance_complete	supported	Models with ¿= 95% local-judge coverage: ['gpt_oss_20b', 'phi4

Table 4: Cross-family selectivity (SSEI) by cache policy. SSEI is safety degradation minus capability degradation relative to the no-cache-policy baseline; positive SSEI indicates safety loss exceeding capability loss. 95% bootstrap CIs in brackets.

Family	**Model**	**sliding window**	**sink+recent**	**policy-pinned**	**user-pinned**	**random matched**
Gemma	Gemma–2–9B–IT	−0.005 [-0.008, -0.003]	−0.001 [-0.003, 0.001]	−0.004 [-0.006, -0.002]	−0.003 [-0.008, 0.001]	−0.000 [-0.003, 0.003]
OpenAI	GPT–OSS–20B	−0.068 [-0.088, -0.048]	−0.068 [-0.088, -0.048]	−0.079 [-0.099, -0.060]	−0.082 [-0.101, -0.063]	−0.079 [-0.099, -0.060]
Llama	Llama–3.1–8B–Instruct	−0.085 [-0.105, -0.067]	0.007 [0.002, 0.012]	−0.001 [-0.007, 0.004]	0.024 [0.016, 0.031]	0.022 [0.014, 0.029]
Mistral	Mistral–7B–Instruct–v0.3	0.009 [0.002, 0.017]	0.001 [-0.002, 0.004]	−0.004 [-0.008, -0.000]	−0.008 [-0.015, -0.002]	0.002 [-0.003, 0.006]
OLMo	OLMo–3–7B–Instruct	−0.004 [-0.008, 0.000]	−0.010 [-0.014, -0.005]	−0.016 [-0.020, -0.011]	0.026 [0.019, 0.033]	0.005 [0.000, 0.010]
Phi	Phi–4	0.084 [0.076, 0.091]	0.001 [-0.000, 0.003]	−0.001 [-0.003, 0.000]	0.055 [0.046, 0.063]	0.044 [0.038, 0.050]
Qwen	Qwen2.5–7B base	–	–	–	–	–
Qwen	Qwen2.5–7B–Instruct	0.011 [0.007, 0.016]	−0.003 [-0.008, 0.001]	−0.001 [-0.006, 0.003]	0.017 [0.011, 0.022]	0.012 [0.007, 0.016]
Qwen	Qwen3–9B	0.092 [0.083, 0.101]	−0.005 [-0.009, -0.001]	0.010 [0.006, 0.015]	−0.002 [-0.006, 0.002]	0.020 [0.014, 0.026]